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Java SE 27 & JDK 27

Class ObjectStreamClass

java.lang.Object
    java.io.ObjectStreamClass

All Implemented Interfaces:

Serializable

```
public final class ObjectStreamClass
extends Object
implements Serializable
```

Serialization's descriptor for classes. It contains the name and serialVersionUID of the class. The ObjectStreamClass for a specific class loaded in this Java VM can be found/created using the lookup method.

The algorithm to compute the SerialVersionUID is described in *Java Object Serialization Specification,* Section 4.6, "Stream Unique Identifiers".

Since:

1.1

External Specifications

Java Object Serialization Specification

See Also:

ObjectStreamField,
Java Object Serialization Specification, Section 4, "Class Descriptors",
Serialized Form

Field Summary

Fields

Modifier and Type	Field	Description
static final ObjectStreamField[]	NO_FIELDS	serialPersistentFields value indicating no serializable fields

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
Class<?>	forClass()	

		Return the class in the local VM that this version is mapped to.
ObjectStreamField	**getField**(String name)	Get the field of this class by name.
ObjectStreamField[]	**getFields**()	Return an array of the fields of this serializable class.
String	**getName**()	Returns the name of the class described by this descriptor.
long	**getSerialVersionUID**()	Return the serialVersionUID for this class.
static ObjectStreamClass	**lookup**(Class<?> cl)	Find the descriptor for a class that can be serialized.
static ObjectStreamClass	**lookupAny**(Class<?> cl)	Returns the descriptor for any class, regardless of whether it implements Serializable.
String	**toString**()	Return a string describing this ObjectStreamClass.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Field Details

NO_FIELDS

```
public static final ObjectStreamField[] NO_FIELDS
```

serialPersistentFields value indicating no serializable fields

Method Details

lookup

```
public static ObjectStreamClass lookup(Class<?> cl)
```

Find the descriptor for a class that can be serialized. Creates an ObjectStreamClass instance if one does not exist yet for class. Null is returned if the specified class does not implement java.io.Serializable or java.io.Externalizable.

Parameters:

`cl` - class for which to get the descriptor

Returns:

the class descriptor for the specified class

lookupAny

```
public static ObjectStreamClass lookupAny(Class<?> cl)
```

Returns the descriptor for any class, regardless of whether it implements `Serializable`.

Parameters:

`cl` - class for which to get the descriptor

Returns:

the class descriptor for the specified class

Since:

1.6

getName

```
public String getName()
```

Returns the name of the class described by this descriptor. This method returns the name of the class in the format that is used by the `Class.getName()` method.

Returns:

a string representing the name of the class

getSerialVersionUID

```
public long getSerialVersionUID()
```

Return the serialVersionUID for this class. The serialVersionUID defines a set of classes all with the same name that have evolved from a common root class

and agree to be serialized and deserialized using a common format. NonSerializable classes have a serialVersionUID of 0L.

Returns:

the SUID of the class described by this descriptor

forClass

```
public Class<?> forClass()
```

Return the class in the local VM that this version is mapped to. Null is returned if there is no corresponding local class.

Returns:

the `Class` instance that this descriptor represents

getFields

```
public ObjectStreamField[] getFields()
```

Return an array of the fields of this serializable class.

Returns:

an array containing an element for each persistent field of this class. Returns an array of length zero if there are no fields.

Since:

1.2

getField

```
public ObjectStreamField getField(String name)
```

Get the field of this class by name.

Parameters:

`name` - the name of the data field to look for

Returns:

The ObjectStreamField object of the named field or null if there is no such named field.

toString

```
public String toString()
```

Return a string describing this ObjectStreamClass.

Overrides:

`toString` in class `Object`

Returns:

a string representation of the object

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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